

Benjamin Shaffrey

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Education

University of Amsterdam, MSc in Mathematics Sept 2023 - current

- **Current Average Grade:** 7.4
- **Coursework:** Dynamical Systems, Ergodic Theory, Quantum Computing, Mathematical Neuroscience, Measure Theoretic Probability, Functional Analysis

University of Amsterdam, MSc in Artificial Intelligence Sept 2021 - August 2025

- **Average Grade:** 7.7
- **Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Information Retrieval, Computer Vision

University of Groningen, BSc in Artificial Intelligence Sept 2017 - August 2020

- **Average Grade:** 8
- **Coursework:** Neural Networks, Reinforcement Learning Practical, Algorithms and Datastructures, Philosophy of Cognitive Science, Probability Theory, Control Theory

Guildhall School of Music and Drama, London, BMus in Classical Piano Performance Sept 2013 - August 2016

- **Grade:** 2nd class honours
- **Coursework:** Solo recital performances, concerto performances, chamber music performances, singer accompaniment, improvisation performances, dissertation on memory in piano playing, dissertation on attention during performances.

Experience

AI Safety Evaluations Engineer, Equistamp Nov 2025 - present

- Implemented automated scorers in Python to validate attack completion in realistic software environments, integrating with PostgreSQL databases and containerised C++ applications.
- Designed and documented attack trajectories to evaluate whether AI monitoring systems can detect subtle attacks, contributing to control research conducted for Redwood Research.
- Collaborated on environment design and security requirement specifications to ensure fair red-blue team evaluations.

Machine Learning and Application Developer, Coders Beyond Borders June 2025 - Nov 2025

- Helped to begin development minimal viable product (MVP) for FactStream AI, a platform that aims to assist migrants and refugees in the Netherlands to better understand their rights and obligations under Dutch law and to be matched with legal experts when they need it.
- Developed application code for MVP.
- Optimised retrieval augmented generation system to answer users' queries faithfully and accurately (hyperparameter optimisation and prompt engineering).
- Deployed system and application on Azure.

Instructor, Clarusway Nov 2024 - Mar 2025

- Gave lectures for online courses in Relational Databases (SQL), Machine Learning, Deep Learning, Natural Language Processing, and ML Deployment.

Software Developer, MediaMarkt July 2021 - Jan 2023

- Automated many reporting tasks through developing executables from python code.
- Automated data gathering from emails, i.e. automatically gathering attachments for specific tasks so as to remove need for manual downloads

- Automated complex calculations needed for more extensive reports and forecasts.

Software Developer, Schut Geometrische Meettechniek

Mar 2021 - July 2021

- Extended object recognition system to deal with more complex objects and object localization.

Software Developer Intern, Schut Geometrische Meettechniek

Nov 2020 - Feb 2021

- Implemented a feature extraction pipeline using elliptical Fourier descriptors on images.
- Built object recognition system using these descriptors. Report can be found [here](#).

Teaching Assistant, University of Groningen

Sept 2018 – Apr 2021

- Gave tutorials and labs and graded homework assignments, labs, and exams (with lecturer supervision and guidance) for courses in computer science, machine learning, reinforcement learning, expert systems, knowledge and agent technology, natural language processing, neuroscience and linguistics.
- Head TA for Introduction to the Brain (Apr 2019 - Jun 2019).
- Created coursework (labs, homework assignment, practice questions) for the courses: Introduction to the brain, Algorithms and Data Structures, Knowledge and Agent Technology.

Publications

Reproducibility study of “Data-Driven Methods for Balancing Fairness and Efficiency in Ride-Pooling”

Apr 2022

Sarah De Boer, Radu Alexandru Cosma, Lukas Knobel, Yeskendir Koishekenov, Benjamin Shaffrey

Accessible at: <https://openreview.net/forum?id=BE3Ms3GXhCF>

Projects

FactStream AI

Application

- A platform that aims to assist migrants and refugees in the Netherlands to better understand their rights and obligations under Dutch law and to be matched with legal experts when they need it.
- Skills: python, javascript, css, html, Azure, RAGflow, prompt engineering, RAGAS, postgresql

Feature Refined LLC Estimation

MSc Thesis Repo

- MSc project to propose method to analyse the developmental trajectory of monosemantic neurons during the course of training.
- Skills: python, singular learning theory, developmental interpretability.

Mathematical Neuroscience Project

MN final project Repo

- Project to replicate the main findings in the paper “High frequency stimulation of the subthalamic nucleus eliminates pathological thalamic rhythmicity in a computational model”.
- Skills: Phase plane analysis, stability analysis, bifurcation theory, network modelling.

Quantum Variational Autoencoders

QVAE Repo

- Project that proposes and evaluates the use of existing algorithms for a quantum speedup in the training of variational autoencoders.
- Skills: Mean-field theory, time complexity analysis, variational inference, quantum rejection sampling.

Reproducibility Study

FACT Repo

- Study to reproduce and extend results in paper “Data-Driven Methods for Balancing Fairness and Efficiency in Ride-Pooling”.
- Skills: python, pytorch, LSTM, reinforcement learning, matching algorithms.

Two-Way Conversion Between Bayesian Networks and Argumentation

BSc Thesis Repo

- BSc project to extend existing mapping from Bayesian networks to argumentation frameworks into a bijective mapping.
- Skills: C++, python, Bayesian networks, Argumentation Frameworks, Logic.

Reversible Jump Markov Chain Monte Carlo Simulation

Project SR Repo

- Project to implement model averaging in Bayesian linear regression applied to two datasets. Model averaging

implemented using reversible jump Markov chain Monte Carlo Simulation.

- Skills: R, Bayesian Linear Regression, Reversible Jump MCMC simulation.

Languages and Tools

Languages: C++, C, Python, Java, R, SQL, JavaScript, Matlab, Mathematica, HTML, CSS

Tools: Microsoft SQL Server, Azure, Streamlit, Sagemaker, Jupyter Notebook, Google Colab, Visual Studio Code, Docker, RAGflow, RAGAS, Git/GitHub, LaTeX, Linux/Unix, Pytorch